SmartBin

- binID: String
- location: String
- zone: String
- fillLevel: float
- capacity: float
- status: BinStatus

+ getFillLevel(): float
+ updateStatus(s: BinStatus): void
+ getBinID(): String
+ getLocation(): String
+ isOverflowing(): boolean

1 1..*

IoTSensor

- sensorID: String
- binID: String
- sensorType: String
- status: SensorStatus
- lastReadingTime: DateTime
- retryCount: int

+ measureFillLevel(): float
+ transmitData(server: CentralServer): voi
+ getStatus(): SensorStatus
+ flagMalfunction(): void
+ resetSensor(): void

generates 0..*

FillLevelRecord

- recordID: String
- binID: String
- sensorID: String
- fillPercentage: float
- timestamp: DateTime
- isArchived: boolean

+ logReading(fill: float): void
+ getHistory(binID: String): List
+ correctReading(val: float): void
+ archiveRecord(): void

Dashboard

- dashboardID: String
- supervisorID: String
- lastUpdated: DateTime
- activeAlerts: int
- totalBins: int

+ displayBinMap(): void
+ updateBinStatus(bin: SmartBin): void
+ showActiveAlerts(): void
+ markSensorOffline(s: IoTSensor): void
+ refresh(): void

<<enumeration>>
BinStatus

ACTIVE
WARNING
CRITICAL
OFFLINE

<<enumeration>>
SensorStatus

ACTIVE
MALFUNCTIONING
OFFLINE

processed by

RouteOptimizer

- routeID: String
- assignedBins: List
- optimizedRoute: List
- status: String

+ receiveAlert(a: Alert): void
+ calculateRoute(): List
+ dispatchRoute(): void
+ updateRoute(): void

Alert

- alertID: String
- binID: String
- alertType: AlertType
- fillLevel: float
- timestamp: DateTime
- status: AlertStatus

+ generateAlert(type: AlertType): void
+ acknowledgeAlert(): void
+ resolveAlert(): void
+ purgeAlert(): void
+ getAlertType(): AlertType

creates 0..1

triggers

updates

triggers 0..*

CentralServer

- serverID: String
- ipAddress: String
- status: String
- lastHeartbeat: DateTime
- warningThreshold: float = 80
- criticalThreshold: float = 100

+ receiveData(data: FillLevelRecord): void
+ validateData(): boolean
+ checkThreshold(fill: float): void
+ triggerRouteOptimizer(): void
+ notifyDashboard(): void

<<enumeration>>
AlertType

WARNING
CRITICAL
MALFUNCTION

MaintenanceReport

- reportID: String
- sensorID: String
- technicianID: String
- issueDescription: String
- repairStatus: RepairStatus
- reportDate: DateTime

+ createReport(sensorID: String): void
+ updateStatus(s: RepairStatus): void
+ viewHistory(): List
+ closeReport(): void